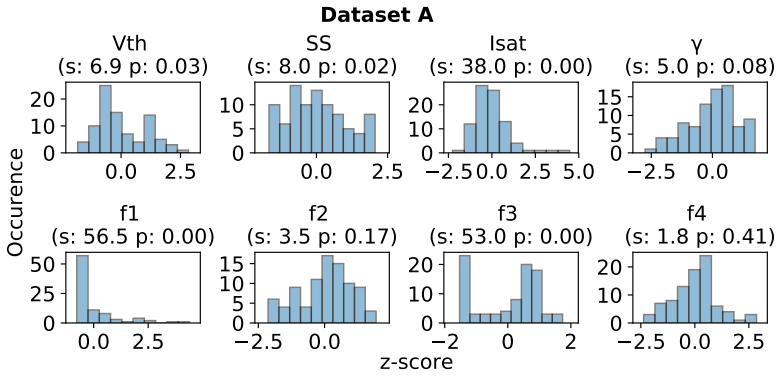
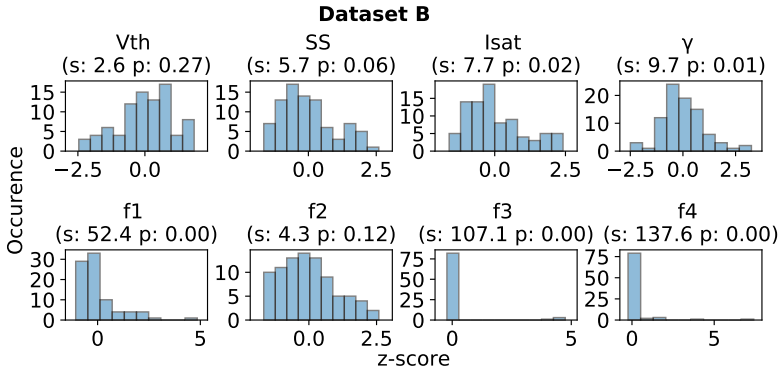
(KS) test [159]. The KS statistic (D) for the difference between the measured and model-generated distribution of a given parameter P is given by Equation 3.13.

$$D = \sup_{P} |F_{model}(P) - F_{measured}(P)| \tag{3.13}$$

where $F_{model}(P)$ and $F_{measured}(P)$ are the distributions of the measured and the model-generated P and *sup* denotes the supremum function.



(a)



(b)

Figure 3.3: The standardized distribution of the parameters. The threshold voltage (V_{th}), sub-threshold slope (SS) and saturation current (I_{sat}) (at $V_{GS} = 1V$) are extracted directly from the characterization data by extrapolating the linear regions of the transistor DC characteristics. Gamma (γ) is an empirically extracted model parameter and f_1, f_2, f_3, f_4 are the fitting parameters of the EGT model. The p and s denote a 2-sided chi squared (χ^2) probability for the hypothesis test and a statistical distance that measures the deviation between the model parameter distribution and a normal distribution, respectively. The z-score is the standard score which is calculated by: $z - score := \frac{x-\mu}{\sigma}$ where x, μ and σ define the observation, sample mean and standard deviation, respectively. The z-score has a $\mu = 0$ and $\sigma = 1$. (a) The Dataset A contain 95 EGTs with channel length (L) and width (W) of $40\mu m$ and of $100\mu m$, respectively (b) The Dataset B contain 88 EGTs with fixed channel length ($L = 40\mu m$) and multiple channel widths ($W = 200, 400, 600$ and $800\mu m$).